Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		ratio C : H ⇒ $\frac{85.6}{12} : \frac{14.4}{1.01}$ ⇒ 7.13 : 14.26 (1) ratio = 1:2 ⇒ empirical formula is CH_2 (1) M_r in the range 50-60 empirical formula M_r = 14 therefore molecular formula must be C_4H_8 (1)		3		3	1	
		(ii)		0.2 mol of C_4H_8 and 0.2 mol of bromine (1) so 1 double bond (1) **must** have some justification		1	1	2		
		(iii)		electrophilic addition	1			1		
		(iv)		radical substitution	1			1		
		(v)		compound **X** $CH_2=CHCH_2CH_3$ or $CH_3CH=CHCH_3$ (1) product of reaction of **X** with bromine in the absence of light $CH_2BrCHBrCH_2CH_3$ or $CH_3CHBrCHBrCH_3$ - must follow from part (i) (1) product of reaction of **X** with excess bromine in sunlight displayed formula of any substituted bromo-compound - must follow from part (ii) (1)			3	3		